US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12416415
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Goldstein; David J.

Hydronic assembly for providing heating, cooling, and ventilation

Abstract

A hydronic assembly connectable to a building's water and air distribution systems to manage and deliver conditioned air to a unit of the building includes an enclosure housing a hydronic coil configured to heat or cool air within the enclosure, a valve control system coupled to the hydronic coil for regulating the flow of water through the hydronic coil, and an air supply booster for mixing recirculated and ventilated air within the enclosure and delivering the same to the unit. The enclosure includes a return air inlet for recirculating air to the enclosure, a supply air outlet for supplying air to the unit, and a return air separation baffle separating the return air inlet air stream from the supply air outlet air stream. The enclosure is removably attachable to the building's air distribution system. The hydronic coil includes piping that is removably attachable to the building's water distribution system.

Inventors: **Goldstein; David J. (Long Island City, NY)**

Applicant: **Hydronic Shell Technologies LLC** (Long Island City, NY)

Family ID: **1000008821490**

Appl. No.: **18/236547**

Filed: **August 22, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230408113 A1	Dec. 21, 2023

Related U.S. Application Data

continuation parent-doc US 17162598 20210129 US 11415328 20220816 child-doc US 17859183
continuation-in-part parent-doc US 17859183 20220707 US 11767990 child-doc US 18236547
us-provisional-application US 62972744 20200211

Publication Classification

Int. Cl.: **F24F1/0057** (20190101); **E04B1/76** (20060101); **E04B2/88** (20060101); **E04F13/00** (20060101); **E04F13/072** (20060101); **E04F13/08** (20060101); **E06B3/66** (20060101); **F24F5/00** (20060101)

U.S. Cl.:

CPC **F24F1/0057** (20190201); **E04B1/7629** (20130101); **E04B1/7645** (20130101); **E04B2/88** (20130101); **E04F13/007** (20130101); **E04F13/072** (20130101); **E06B3/66** (20130101); **F24F5/0075** (20130101); E04F13/0833 (20130101); E04F13/0869 (20130101); E04F2290/023 (20130101); F24F2005/0082 (20130101); F24F2221/17 (20130101); F24F2221/20 (20130101); F24F2221/50 (20130101); Y02B30/00 (20130101)

Field of Classification Search

CPC: F24F (1/0057); F24F (5/0075); F24F (2005/0082); F24F (2221/17); E04B (1/7629); E04B (2/88); E04F (13/0869)

USPC: 165/48.1

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4296798	12/1980	Schramm	N/A	N/A
4593511	12/1985	Hakasaari	N/A	N/A
5347779	12/1993	Jordan	N/A	N/A
5497826	12/1995	Ingram	N/A	N/A
5566525	12/1995	Schwarz	N/A	N/A
6754997	12/2003	Bonin	N/A	N/A
6805298	12/2003	Corbett	N/A	N/A
6860073	12/2004	Chien	N/A	N/A
8621787	12/2013	Barry et al.	N/A	N/A
8978324	12/2014	Collins	52/404.1	E04C 2/521
9182133	12/2014	Weber	N/A	N/A
9273463	12/2015	Kossman	N/A	F24F 11/74
9382706	12/2015	Warren	N/A	N/A
9988814	12/2017	Schluter	N/A	N/A
10179996	12/2018	Schluter	N/A	N/A
11415328	12/2021	Goldstein	N/A	N/A
11821649	12/2022	Roj	N/A	N/A
2004/0154245	12/2003	Chien	N/A	N/A
2005/0040152	12/2004	Koschenz et al.	N/A	N/A
2013/0180191	12/2012	Teron	N/A	N/A
2015/0284944	12/2014	Warren	N/A	N/A
2016/0186431	12/2015	Schluter	N/A	N/A
2017/0254550	12/2016	Poccianti	N/A	F24F 7/08
2018/0245342	12/2017	Schluter	N/A	N/A

2019/0093356	12/2018	Collins et al.	N/A	N/A
2019/0301156	12/2018	Sandefur	N/A	N/A
2020/0232663	12/2019	Buff	N/A	N/A
2020/0392738	12/2019	Passon et al.	N/A	N/A
2021/0108825	12/2020	Roj	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2932628	12/1980	DE	N/A
19521414	12/1995	DE	N/A
2866040	12/2004	FR	N/A
8004182	12/1979	NL	N/A
WO2004106811	12/2003	WO	N/A
2013/121044	12/2012	WO	N/A
WO2016000089	12/2015	WO	N/A
WO2017156016	12/2016	WO	N/A

OTHER PUBLICATIONS

Written Opinion of the International Searching Authority for PCT/US2021/016026 established by the ISA/US completed on Apr. 26, 2021. cited by applicant
International Preliminary Report on Patentability of the International Preliminary Examining Authority for PCT/US2021/016026 established by the IPEA/US completed on Dec. 15, 2021. cited by applicant

---

Primary Examiner: Tanenbaum; Steve S

Attorney, Agent or Firm: Goldstein Law Offices, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/859,183, filed on Jul. 7, 2022, which is a continuation of U.S. patent application Ser. No. 17/162,598 (now issued U.S. Pat. No. 11,415,328), filed on Jan. 29, 2021, which claims priority to U.S. Provisional Patent Application Ser. No. 62/972,744, filed on Feb. 11, 2020, which are relied upon and incorporated herein by reference in their entirety. The entire disclosure of any publication or patent document mentioned herein is entirely incorporated by reference.

TECHNICAL FIELD
(1) The present disclosure relates generally to hydronic systems for providing, heating, cooling, and ventilation to new or existing buildings. More particularly, the present disclosure relates to an insulated facade panel conditioning system including modular panels configured for installation on new or existing buildings such as multifamily residential buildings, condominiums, hotels, or dormitories to create an insulated shell therearound. The insulated facade panel conditioning system includes integrated heating, ventilation, and air conditioning (HVAC) piping and ductwork within each modular panel that distributes highly efficient heating and cooling to individual units within these buildings.
BACKGROUND
(2) Buildings are a major contributor to global energy consumption and greenhouse gas emissions. There is urgent need to reduce these emissions especially within the older existing building stock,

such as multifamily residential apartment buildings and condominiums. By way of example, New York City has passed the Climate Mobilization Act, which requires that large buildings comply with emissions limits. However, the industry currently lacks a comprehensive and cost-effective method of implementing retrofits to existing buildings that drives them towards net-zero energy consumption. Moreover, because a significant portion of existing buildings are older, they are in much need of revitalization to their appearance and infrastructure.

(3) Accordingly, there is a need for a facade panel conditioning system that provides a new cost-effective approach to greatly reducing energy consumption and greenhouse gas emissions while minimizing disruption to existing tenants and/or occupants. Moreover, there is a need for a facade panel conditioning system that once installed provides the additional benefits of an updated HVAC infrastructure offering improved comfort and air quality and simplified operation and maintenance, as well as a new facade appearance that revitalizes our existing buildings and neighborhoods.

(4) While these units may be suitable for the particular purpose employed, or for general use, they would not be as suitable for the purposes of the present disclosure as disclosed hereafter.

(5) In the present disclosure, where a document, act or item of knowledge is referred to or discussed, this reference or discussion is not an admission that the document, act or item of knowledge or any combination thereof was at the priority date, publicly available, known to the public, part of common general knowledge or otherwise constitutes prior art under the applicable statutory provisions, or is known to be relevant to an attempt to solve any problem with which the present disclosure is concerned.

(6) While certain aspects of conventional technologies have been discussed to facilitate the present disclosure, no technical aspects are disclaimed and it is contemplated that the claims may encompass one or more of the conventional technical aspects discussed herein.

BRIEF SUMMARY

(7) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system capable of being installed on a new or existing building. Accordingly, the present disclosure provides a facade panel conditioning system comprising modular panels including connection anchors that connect to the structure of a building.

(8) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system capable of being installed on a new or existing building and creates an insulated shell around the building. Accordingly, the present disclosure provides a facade panel conditioning system comprising modular panels that connect to one another over the exterior of a building in an airtight, watertight, and vapor-tight fashion.

(9) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system that eliminates the need for the combustion of fossil fuels, thereby reducing greenhouse gas emissions. Accordingly, the present disclosure provides a facade panel conditioning system including an HVAC system that is powered by electricity and/or integrates with a building's electrical infrastructure.

(10) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system that reduces the heating and cooling loads required to heat and cool new or existing buildings, thereby making the building more energy efficient. Accordingly, the present disclosure provides a facade panel conditioning system including high-performance insulated modular panels that in combination increase the R-value of the insulation of the new or existing building, thereby requiring a smaller HVAC system/piping and ductwork to heat and cool the building efficiently.

(11) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system that enables the addition or installation of new HVAC piping and ductwork to the infrastructure of a new or existing building. Accordingly, the present disclosure provides a facade panel conditioning system including modular panels that are configured to create an air cavity between the modular panel and the facade of the building once installed, so as to enable new

HVAC piping and ductwork to be positioned in the air cavity and distribute heating or cooling water and air thereto.
(12) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system capable of distributing heat, cooling, and ventilation to each individual unit of a new or existing building. Accordingly, the present disclosure provides a facade panel conditioning system including a modular panel including HVAC piping and ductwork circulation units, which distribute heating, cooling, and ventilation to the individual units of the building.
(13) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system capable of distributing heat, cooling, and ventilated air throughout the entirety of a new or existing building. Accordingly, the present disclosure provides a facade panel conditioning system including modular panels that have HVAC piping and ductwork that is connectable between each modular panel, thereby enabling the distribution of heating, cooling, and ventilated air throughout the building once the facade panel conditioning system is installed over the building.
(14) An aspect of an example embodiment in the present disclosure is to provide a facade panel conditioning system that once installed over a new or existing building prevents the migration of air, fire, and smoke horizontally or vertically from the air cavity of a modular panel to the air cavity of an adjacent modular panel. Accordingly, the present disclosure provides a facade panel conditioning system including isolation baffles separating the air cavities and HVAC piping and ductwork of adjacent modular panels.
(15) The present disclosure addresses at least one of the foregoing disadvantages. However, it is contemplated that the present disclosure may prove useful in addressing other problems and deficiencies in a number of technical areas. Therefore, the claims should not necessarily be construed as limited to addressing any of the particular problems or deficiencies discussed hereinabove. To the accomplishment of the above, this disclosure may be embodied in the form illustrated in the accompanying drawings. Attention is called to the fact, however, that the drawings are illustrative only. Variations are contemplated as being part of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the drawings, like elements are depicted by like reference numerals. The drawings are briefly described as follows.
(2) FIG. **1** is a perspective view of the facade panel conditioning system partially installed over the exterior of a building, illustrating the modular installation of the facade panel conditioning system according to one embodiment of the present disclosure.
(3) FIG. **2** is a perspective close-up and partial phantom view of the facade panel conditioning system partially installed over the exterior of a building, illustrating the configuration of the HVAC piping and ductwork of the facade panel conditioning system within each modular panel and the interconnection of the HVAC piping and ductwork after installation of the modular panels according to one embodiment of the present disclosure.
(4) FIG. **3** is an elevation view of the interior of the modular panels of the facade panel conditioning system after full installation of the facade panel conditioning system on a building, illustrating the distribution of the HVAC piping and ductwork of the facade panel conditioning system throughout the building and to the individual units of the building according to one embodiment of the present disclosure.
(5) FIG. **4** is a plan view of the roof of a building in which the facade panel conditioning system has been fully installed on a building, illustrating the HVAC plant and the distribution of the HVAC piping and ductwork on the roof according to one embodiment of the present disclosure.

(6) FIG. **5** is a cross-sectional view of a building in which the facade panel conditioning system has been fully installed on a building, illustrating a cross-section of the modular panels of the facade panel conditioning system along the exterior of the building according to one embodiment of the present disclosure.
(7) FIG. **6** is a schematic view of an assembled facade panel conditioning system, illustrating the distribution of the HVAC piping and ductwork of the facade panel conditioning system throughout the building and to the individual units of the building according to one embodiment of the present disclosure.
(8) FIG. **7** is a cross-sectional view of a building in which the facade panel conditioning system has been fully installed on a building, illustrating a cross-section of the modular panels of the facade panel conditioning system along the exterior of the building and the air duct ventilation system formed by the network of modular facade panels installed over the exterior of the building according to one embodiment of the present disclosure.
(9) FIG. **8**A is a vertical cross-sectional view of a modular panel of the facade system attached to the exterior of a building, illustrating the configuration of the components of the modular panel with respect to the exterior of a building when the modular panel is attached to the exterior of the building according to one embodiment of the present disclosure.
(10) FIG. **8**B is an elevation view of an interior of a modular panel of the facade panel conditioning system, illustrating the configuration of the components of the modular panel according to one embodiment of the present disclosure.
(11) FIG. **8**C is a horizontal cross-sectional view of the modular panel of the facade panel conditioning system attached to the exterior of a building, illustrating the configuration of the components of the modular panel within the air cavity, which is formed between the modular panel and the exterior of a building when the modular panel is attached to the exterior of the building according to one embodiment of the present disclosure.
(12) FIG. **9** is a perspective view of a modular panel of the facade panel conditioning system, illustrating the configuration of the components of the modular panel according to one embodiment of the present disclosure.
(13) FIG. **10** is a perspective view of a modular panel of the facade panel conditioning system, illustrating the configuration of the components of the modular panel according to another embodiment of the present disclosure.
(14) FIG. **11** is a cross-sectional view of an apartment unit of a building having a modular panel of the facade panel conditioning system mounted to the exterior of the building, illustrating the relative orientation and positioning of the panel and its components with respect to the exterior of the building as well as the way in which heated or cooled air is distributed to the apartment unit according to one embodiment of the present disclosure.
(15) FIG. **12** is an elevation view of several different panel configurations of the modular panel of the facade panel conditioning system, illustrating the components of each individual modular panel of each of the panel configurations and the relative orientation of each modular panel and its components with respect to adjacent modular panels when modularly attached to the exterior of a building according to one embodiment of the present disclosure.
(16) FIG. **13**A is an elevation view of an interior of a modular panel of the facade panel conditioning system, illustrating the components of the hydronic assembly and the configuration of the hydronic assembly with respect to the hydronic piping and air duct of the modular panel according to one embodiment of the present disclosure.
(17) FIG. **13**B is a front elevation schematic view of the interior of the hydronic assembly of FIG. **13**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(18) FIG. **13**C is a side schematic view of the interior of the hydronic assembly of FIG. **13**A, illustrating the configuration of the components of the hydronic assembly according to one

embodiment of the present disclosure.
(19) FIG. **13**D is a top schematic view of the interior of the hydronic assembly of FIG. **13**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(20) FIG. **13**E is a side view of the hydronic assembly of FIG. **13**A, illustrating the casing angles of the hydronic assembly according to one embodiment of the present disclosure.
(21) FIG. **14**A is an elevation view of an interior of a modular panel of the facade panel conditioning system, illustrating the components of the hydronic assembly and the configuration of the hydronic assembly with respect to the hydronic piping and air duct of the modular panel according to another embodiment of the present disclosure.
(22) FIG. **14**B is a front elevation schematic view of the interior of the hydronic assembly of FIG. **14**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(23) FIG. **14**C is a side schematic view of the interior of the hydronic assembly of FIG. **14**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(24) FIG. **14**D is a top schematic view of the interior of the hydronic assembly of FIG. **14**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(25) FIG. **14**E is a side view of the hydronic assembly of FIG. **14**A, illustrating the casing angles of the hydronic assembly according to one embodiment of the present disclosure.
(26) FIG. **15**A is an elevation view of an interior of a modular panel of the facade panel conditioning system, illustrating the components of the hydronic assembly and the configuration of the hydronic assembly with respect to the hydronic piping and air duct of the modular panel according to another embodiment of the present disclosure.
(27) FIG. **15**B is a front elevation schematic view of the interior of the hydronic assembly of FIG. **15**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(28) FIG. **15**C is a side schematic view of the interior of the hydronic assembly of FIG. **15**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(29) FIG. **15**D is a top schematic view of the interior of the hydronic assembly of FIG. **15**A, illustrating the configuration of the components of the hydronic assembly according to one embodiment of the present disclosure.
(30) The present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, which show various example embodiments. However, the present disclosure may be embodied in many different forms and should not be construed as limited to the example embodiments set forth herein. Rather, these example embodiments are provided so that the present disclosure is thorough, complete and fully conveys the scope of the present disclosure to those skilled in the art.

DETAILED DESCRIPTION

(31) FIG. **1** illustrates a facade panel conditioning system **10** for installation on the exterior **12** of a new or existing building **14**, such as a new construction or an existing multifamily residential building, condominium, hotel, or dormitory. The facade panel conditioning system **10** includes a plurality of modular panels **16** that are each configured to attach to the structure of the building **14** as well as modularly attach to one another over the exterior **12** of the building **14**. The exterior **12** of the building **14** includes portions that correspond to the individual units within the building **14**. The plurality of modular panels **16** are configured to attach to and cover these portions of the exterior **12** such that each of the plurality of modular panels **16** corresponds to an individual unit of the building **14**. When attached to the facade and to one another over the exterior **12**, the plurality

of modular panels **16** form an insulated shell around the building **14** that encloses or envelopes the exterior **12**.
(32) In embodiments, each of the modular panels of the plurality of panels **16** is fabricated to include the same dimension as the interior wall of the individual unit, such that a modular panel attached to the exterior **12** spans or is coextensive with the interior wall of the individual unit of the building **14**. In some embodiments, the facade panel conditioning system **10** comprises a gasket or sealant (not shown) configured to seal the plurality of modular panels **16** to one another when attached to the building **14**. The sealant creates an airtight and watertight seal between an interior of the plurality of modular panels **16** and an exterior of the plurality of modular panels **16**, thereby making the insulated shell around the building airtight and watertight.
(33) Referring now to FIG. **2**, each of the plurality of modular panels **16** comprises insulation **18**, a window assembly **20**, hydronic piping **22**, and an air duct **24**. The hydronic piping **22**A of a modular panel **16**A of the plurality of modular panels **16** connects to the hydronic piping **22**B of an adjacent modular panel **16**B of the plurality of modular panels **16**. In embodiments, the adjacent modular panel **16**B is a modular panel attached to the individual unit of the building **14** that is immediately above the individual unit of the building to which the modular panel **16**A has been attached. In embodiments, the hydronic piping **22**A of modular panel **16**A attaches to the hydronic piping **22**B of adjacent modular panel **16**B after the plurality of modular panels **16** are attached to the structure of the building **14**. In embodiments, the hydronic piping **22**A of modular panel **16**A attaches to the hydronic piping **22**B of adjacent modular panel **16**B before the plurality of modular panels **16** are attached to the structure of the building **14**. In embodiments, the hydronic piping **22**A of modular panel **16**A attaches to the hydronic piping **22**B of adjacent modular panel **16**B after the plurality of modular panels **16** are attached to the structure of the building **14** but before the plurality of modular panels **16** are sealed.
(34) The air duct **24**A of modular panel **16**A connects to the air duct **24**B of an adjacent modular panel **16**B. In embodiments, the air duct **24**A of modular panel **16**A attaches to the air duct **24**B of adjacent modular panel **16**B after the plurality of modular panels **16** are attached to the structure of the building **14**. In some embodiments, the air duct **24**A of modular panel **16**A attaches to the air duct **24**B of adjacent modular panel **16**B before the plurality of modular panels **16** are attached to the structure of the building **14**. In other embodiments, the air duct **24**A of modular panel **16**A attaches to the air duct **24**B of adjacent modular panel **16**B after the plurality of modular panels **16** are attached to the structure of the building **14** but before the plurality of modular panels **16** are sealed.
(35) Referring now to FIG. **3**, the hydronic piping **22**A, **22**B of the modular panels **16**A, **16**B attach to one another to form a hydronic piping system **22**C that distributes heating or cooling water throughout the insulated shell and building **14**. In embodiments, the hydronic piping system **22**C extends vertically through the facade panel conditioning system **10**. The hydronic piping system **22**C comprises a network of hydronic piping formed from the interconnection of the hydronic piping **22** of adjacent modular panels **16**.
(36) The air ducts **24**A, **24**B of the modular panels **16**A, **16**B attach to one another to form an air duct ventilation system **24**C that distributes air throughout the insulated shell and building **14**. In embodiments, the air duct ventilation system **24**C extends vertically through the facade panel conditioning system **10**. The air duct ventilation system **24**C comprises a network of air ducts formed from the interconnection of the air ducts **24** of adjacent modular panels **16**.
(37) Referring now to FIG. **4**, the facade panel conditioning system **10** comprises a hydronic heat pump system **26** to operate the hydronic piping system **22**C. In embodiments, the hydronic heat pump system **26** is connected to the hydronic piping system **22**C at the roof **30** of the building **14**. In some embodiments, the hydronic heat pump system **26** is retrofit, installed, or attached to the roof **30** of the building **14** such that it is attached exteriorly to the building **14**. The hydronic heat pump system **26** may also be installed within the interior of the building **14** as such as in a

mechanical or utility room. The hydronic heat pump system **26** is electric thereby eliminating the combustion of fossil fuels for heating and cooling. In operation, heated and chilled water provided to the hydronic piping system **22**C is generated by the hydronic heat pump system **26**. The hydronic heat pump system **26** may comprise air or water-source heat pumps and distribution pumps. In one embodiment, the hydronic heat pump system **26** is factory assembled for delivery to the project site in a single module. The water temperatures required to provide heating and cooling throughout the insulated shell are moderate, allowing for extremely efficient operation of the hydronic heat pump system **26** throughout the year.

(38) In embodiments, the facade panel conditioning system **10** comprises a ventilation air handling unit **28** to operate the air duct ventilation system **24**C. In embodiments, the ventilation air handling unit **28** is connected to the air duct ventilation system **24**C at the roof **30** of the building **14**. In some embodiments, the ventilation air handling unit **28** is retrofit, installed, or attached to the roof **30** of the building **14** such that it is disposed exteriorly to the building **14**. The ventilation air handling unit **28** may also be installed within the interior of the building **14** as such as in a mechanical or utility room.

(39) In operation, the ventilation air handling unit **28** conditions air and supplies the air to the air duct ventilation system **24**C. In embodiments, the ventilation air handling unit **28** comprises a heat pump air handling unit. The ventilation air handling unit **28** provides a high level of filtration to ensure superior air quality throughout the insulated shell and within each apartment or unit. Depending on outdoor conditions the ventilation air is cooled or heated by the ventilation air handling unit **28** so as to be distributed at a neutral temperature. During cooling the ventilation air handling unit **28** dehumidifies ventilated air to a very low dewpoint by mechanical sub-cooling with reheat or by desiccant dehumidification. In some embodiments, the ventilation air handling unit **28** comprises an enthalpy recovery heat exchanger for receiving exhaust air therethrough to precondition the air entering from the outside and thereby reducing the conditioning load and energy consumption at the ventilation air handling unit **28**.

(40) Referring now to FIG. **5**, each of the plurality of modular panels **16** of the facade panel conditioning system **10** is in communication with the hydronic heat pump system **26**. The hydronic heat pump system **26** includes a water circulation pump **27** for circulating water throughout the hydronic heat pump system **26**. The comprises an anchor **35** which fixedly attaches each of the plurality of modular panels **16** to the structure **37** of the building **14**. The structure **37** comprises any structural component of the building **14** such as a structural slab, beam, and column. When installed over an existing facade, the anchor **35** attaches to the structure **37** of the building **14** such that the panel juts out from the exterior **12**, thereby forming an air cavity **34** between the plurality of modular panels **16** and the exterior **12** of the building **14**.

(41) In embodiments, the air cavity **34** is 6 to 12 inches in width, i.e., the distance between the wall of the exterior **12** of the building **14** and the modular panel is 6 to 12 inches. In some embodiments, the air cavity **34** is 6 to 8 inches in width. In other embodiments, the air cavity **34** is 10 to 12 inches in width. In embodiments, the insulated shell **32** is completed at the roof **30**. The insulated shell **32** extends vertically past the roof **30** to form a new parapet **36** having insulation that fully encloses and insulates the shell **32** on the exterior **12**.

(42) In embodiments, when installed on an existing building, once the insulated shell **32** is formed around the building **14**, a user removes the existing windows of the individual units **33** of the building **14** to receive heat and air from the hydronic piping system **22**C and the air duct ventilation system **24**C within the shell **32**. Next, a user optionally removes the existing insulation of the wall **38** of the individual units **33** of the building **14** that corresponds to the facade of the individual unit **33** to maximize heat transfer from the hydronic piping system **22**C to the air cavity **34** and individual unit **33**. Next, the user seals off the air cavities **34** from the individual units **33**, for example, by sealing the window opening, formed by removing the windows, with gypsum board or similar material. Finally, a user optionally removes any of the existing HVAC system components

such as piping, radiators and air conditioning units from the individual unit **33**.
(43) Referring now to FIG. **6** and FIG. **7**, the air duct ventilation system **24**C is coupled to the ventilation air handling unit **28** and includes an air supply riser **38** and an air exhaust riser **40**. While the airtight and watertight seal of the insulated shell creates the desired effect of reducing heating and cooling loads and improved comfort, the seal necessitates the provision of mechanically supplied ventilation air to maintain excellent indoor air quality within the individual units. Accordingly, the supply riser **38** distributes air **41** to each of the individual units **33** corresponding to each of the plurality of modular panels **16**, while the exhaust riser **40** removes exhaust air **43** from each of the individual units **33** to ensure a neutral or slightly positive air pressure. The quantity of air required at each apartment is low enough that the air duct ventilation system **24**C can be positioned within the air cavity.
(44) Referring now to FIGS. **8**A, FIG. **8**B, and FIG. **8**C, each of the plurality of modular panels **16** comprises an interior surface **42**, an exterior surface **44**, a first side **46**, a second side **48**, an upper end **50**, and a lower end **52**. The first side **46** and the second side **48** are separated by the window assembly **20**. The upper end **50** and the lower end **52** are also separated by the window assembly **20**. The window assembly **20** may comprise one or more separate and distinct windows. In embodiments, the window assembly **20** is disposed centrally along a longitudinal axis of the plurality of modular panels **16** and offset toward the upper end **50** of the plurality of modular panels **16**. In some embodiments, the window assembly **20** comprises a triple pane window to increase an insulation R-value of the window assembly **20** of each of the plurality of modular panels **16**. In embodiments, the window assembly comprises a window that produces an R-value of 3 ft.sup.2.Math.° F..Math.h/BTU to 10 ft.sup.2.Math.° F..Math.h/BTU. The window assembly includes an upper end **20**A, a lower end **20**B, a first side **20**C, and a second side **20**D.
(45) The lower end **52** of each of the plurality of modular panels **16** defines an interior volume comprising the insulation **18**. In embodiments, the insulation **18** comprises any known insulative material or combination of insulative materials and insulative layering that produces an R-value of 10 ft.sup.2.Math.° F..Math.h/BTU to 40 ft.sup.2.Math.° F..Math.h/BTU. For example, in one embodiment, the insulation **18** comprises a foam layered between a metallic panel. In embodiments, the plurality of modular panels **16** are prefabricated, enabling delivery directly to a project site for quick installation and minimal disruption to tenants or owners within the building. Each of the plurality of modular panels **16** may be manufactured in various configurations including different shapes and sizes so as to be installed on any new or existing building facade. In some embodiments, the exterior surface **44** may include any variety of known materials, patterns, and/or colors to allow for a wide range of designs on the exterior surface for architectural expression. For example, in embodiments, the exterior surface may comprise, wood planks, sculpted fiber glass, metal panels, cement board, molded polycarbonate, fiberglass, or polycarbonate. In other embodiments, the exterior surface **44** comprises exterior window shading, lighting, and/or building integrated photovoltaics.
(46) The air cavity **34** comprises a first side **53**, a second side **55**, an upper end **57**, and a lower end **59**. The first side **53** and the second side **55** are separated by the window assembly **20**. The upper end **57** and the lower end **59** are also separated by the window assembly **20**. In embodiments, the air cavity **34** is utilized for routing other utilities such as low voltage communication wiring or power conduits.
(47) The hydronic piping **22** and the air duct **24** of each of the modular panels of the plurality of modular panels **16** are disposed within the air cavity **34**. The hydronic piping **22** comprises a supply riser **54**, a return riser **56**, and air cavity supply piping **58**. The supply riser **54** and the return riser **56** are part of the larger hydronic piping system and configured to deliver heat throughout the insulated shell and building **14**. The air cavity supply piping **58** is configured to transfer heat directly to each respective individual air cavity **34**. In embodiments, the hydronic piping **22** is uninsulated to allow for heat transfer directly to the air cavity **34**.

(48) In embodiments, the supply riser **54** and the return riser **56** are positioned within the first side **53** of the air cavity **34** adjacent to the first side **46** of the plurality of modular panels **16**. The air cavity supply piping **58** is positioned within the lower end **59** of the air cavity **34** adjacent to the lower end **52** of the plurality of modular panels **16**. The air cavity supply piping **58** is coplanar with the supply riser **54** and the return riser **56**. The air cavity supply piping **58** extends outwardly from the supply riser **54** into a first area of the lower end **59** of the air cavity **34**, through the lower end **59** of the air cavity **34**, and back to the return riser **56** from a second area of the lower end **59** of the air cavity **34**. The air cavity supply piping **58** is arranged in a series of rows so as to increase the area of the air cavity supply piping **58** within the air cavity **34**. In some embodiments, the air cavity supply piping **58** is attached to the interior surface **42** of the plurality of modular panels **16** at the lower end **52** of the plurality of modular panels **16**. In other embodiments, the air cavity supply piping **58** comprises a finned surface **61** to allow for additional heat transfer to the air cavity **34**.
(49) In operation, as heated water circulates through the hydronic piping **22** the air within the air cavity **34** rises in temperature. The rise in temperature within the air cavity **34** subsequently raises the temperature of the interior wall of the unit **33** to act as a radiant heating surface to the interior of the unit **33**. During cooling, chilled water produces a similar radiant cooling effect.
(50) The air duct **24** comprises an air supply/exhaust riser **60** and an air supply/exhaust branch duct **62**. In some embodiments, the air supply/exhaust riser **60** is an air supply riser or an air exhaust riser. The air supply/exhaust branch duct **62** is configured to deliver a ventilated air stream directly to the air cavity **34**.
(51) In embodiments, the air supply/exhaust riser **60** is positioned within the second side **55** of the air cavity **34** adjacent to the second side **48** of the plurality of modular panels **16**. In some embodiments, the air supply/exhaust branch duct **62** is positioned within the lower end **59** of the air cavity **34** adjacent to the lower end **52** of the plurality of modular panels **16**. In other embodiments, the air cavity supply/exhaust air ventilation branch duct **62** is positioned within the second side **55** of the plurality of modular panels **16** (see FIGS. **14**A-**14**D and FIGS. **15**A-**15**D). The branch duct **62** is coplanar with the air supply/exhaust riser **60**. The branch duct **62** may extend outwardly or perpendicularly from the air supply/exhaust riser **60** into the lower end **59** or the second side of the air cavity **34** toward the window assembly **20**. In some embodiments, the branch duct **62** may extend to and be coupled with the window assembly **20** such that the branch duct **62** may deliver ventilated air directly to a unit **33** (see FIG. **14**A). The window assembly **20** may include an access door **23** adjacent to the branch duct **62** to access the branch duct **62** for maintenance (see FIG. **13**A, FIG. **14**A, and FIG. **15**A). In embodiments, the access door **23** is disposed at either the upper end **20**A or the lower end **20**B of the window assembly **20** (see FIG. **13**A). In some embodiments, the access door **23** is disposed at either the first side **20**C or the second side **20**D of the window assembly **20** (see FIGS. **14**A and **15**A). In embodiments, the supply riser **54**, the return riser **56**, and the air supply/exhaust riser **60** extend parallel relative to each other within the air cavity **34**. In some embodiments, the air supply/exhaust branch duct **62** comprises a portion of the air cavity supply piping **58** within an interior of the branch duct **62** to provide additional heating or cooling capacity to the ventilated air stream. In some embodiments, the air supply/exhaust branch duct **62** comprises a balancing damper **64** to regulate the ventilated air stream to the air cavity **34** and throughout the insulated shell. In embodiments, the branch duct **62** comprises an air cavity supply branch **63** for delivering a small amount of ventilated air directly to the air cavity **34**. The air cavity supply branch **63** may be an orifice or a vent disposed on the branch duct **62**, or ductwork extending outwardly from the branch duct **62**. The air cavity supply branch **63** may include an air cavity supply branch valve **67** (see FIG. **13**A, FIG. **14**A, and FIG. **15**A) for controlling the flow of air from the air cavity supply branch **63** to the air cavity **34**. The air cavity supply branch valve **67** may completely shut-off the flow of air through the supply branch **63** or control the quantity of air being dispersed into the air cavity **34** through the supply branch **63**.
(52) In embodiments, each of the plurality of modular panels **16** also comprises a hydronic

assembly **68** including an enclosure **69** having a hydronic coil **65**, an air supply diffuser **70** including return air inlets **79**, an air supply booster fan **71**, a return air separation baffle **73**, and an access gate **75**. The hydronic assembly **68** protrudes outwardly from the lower end **66** of the window assembly **20**. In embodiments, the hydronic assembly **68** is attached to the lower end **66**, immediately below the window of the window assembly **20**, serving as a new windowsill or part of the old windowsill after installation. In other embodiments, the hydronic assembly **68** can be oriented vertically and attached to either side of the window assembly **20**. The heat and air from the hydronic piping **22** and the air duct **24** enter the individual units **33** through the hydronic assembly **68**. The hydronic assembly **68** is coupled to the air supply/exhaust branch duct **62** to broaden the air supply from the air supply/exhaust branch duct **62** to the individual units **33**. In embodiments, the hydronic piping **22** comprises a valve **39** disposed within the hydronic assembly **68** to adjust the water flow of the hydronic piping **22** to provide the required amount of heating or cooling as indicated by a thermostat within an individual unit **33**. The valve **39** is located such that it is easily accessible from within the individual unit **33**. In some embodiments, the valve **39** is coupled to the air cavity supply piping **58** or the hydronic coil **65** and accessible via the access gate **75** in the hydronic assembly **68**. The air supply booster fan **71** may either be a linear fan or an axial fan.
(53) In embodiments, the hydronic coil **65** is piped in series, or forms a part of, the air cavity supply piping **58** to provide additional heating or cooling capacity to the ventilated air stream. For example, in embodiments, the air cavity supply piping **58** extends outwardly from the supply riser **54** directly into the hydronic assembly **68** and then out of the hydronic assembly **68** to the first area of the lower end **59** of the air cavity **34**. Note, the portion of the air cavity supply piping **58** that is disposed within the hydronic assembly **68**, or extends through the hydronic assembly **68**, is the hydronic coil **65**. The hydronic coil **65** is arranged in a series of rows so as to increase the area of the hydronic coil within the hydronic assembly **68**. The hydronic coil **65** comprises a finned surface **61** to allow for additional heat transfer to the enclosure **69**. In embodiments, the plurality of panels **16** also comprise condensate disposal risers or pumps for removing condensation formed in the hydronic assembly **68** and/or hydronic coil **65**. In other embodiments, the system does not form condensate and no means of condensate disposal are required.
(54) In embodiments, the hydronic piping **22** includes a single supply riser **54** and a single return riser **56** each circulating either heated or chilled water. In other embodiments, the hydronic piping includes two supply risers **54** and two return risers **56**, wherein one of the supplies risers **54** or the return risers **56** is dedicated to heated water and the other is dedicated to chilled water, and includes valves within the hydronic assembly **68** to select between the heated water and the chilled water.
(55) In operation, capacity can be enhanced by increasing the air flow of the ventilated air stream over the hydronic coil **65** by inducing recirculated room air via the air supply booster fan **71** to mix with the ventilated air stream prior to contact with the hydronic coil **65**.
(56) Referring now to FIGS. **13**A-**13**E, FIGS. **14**A-**14**E, and FIGS. **15**A-**15**D in conjunction with FIG. **8**B and FIG. **3**, in embodiments, the hydronic assembly **68** is disposed adjacent to the window assembly **20** and protrudes outwardly with respect thereto. The hydronic assembly **68** may be mounted to either the upper end **20**A, the lower end **20**B, the first side **20**C, or the second side **20**D of the window assembly **20** in order to accommodate the particular configuration of the modular panel to which it pertains or is designed to serve. The hydronic assembly **68** connects to the building's **10** water distribution system or hydronic piping system **22**C via the hydronic piping **22** and to the building's **10** air distribution system or the air duct ventilation system **24**C via the air duct **24** to provide heating, cooling, and ventilation to the particular air cavity **34** or space in which it is disposed and defined to serve. In some embodiments, the hydronic assembly **68** is positioned immediately below the window of the window assembly **20**, serving as a new windowsill or part of the old windowsill after installation. In other embodiments, the hydronic assembly **68** is configurable on a side of the window of the window assembly **20**.
(57) In embodiments, the hydronic assembly **68** includes an enclosure **100** removably attachable to

the hydronic piping **22** and the air duct **24**, a hydronic coil **102** disposed within the enclosure **100** configured to provide heating and cooling capacity to air recirculated or ventilated into the enclosure **100**, a valve control station **104** for control the flow of water through the hydronic coil **102**, and an air supply booster fan **106** configured to increase efficiency and capacity of the hydronic assembly **68**. The enclosure **100** is removably attachable to the hydronic piping **22** via the hydronic supply piping **54**A and hydronic return piping **56**A, which branch off the supply riser **54** and return riser **56**, respectively. The enclosure **100** is attachable to the air duct **24** via the air cavity supply/exhaust air ventilation branch duct **62**.
(58) The enclosure **100** includes a first end **100**A, a second end **100**B, a first side **1000**, a second side **100**D, a first sidewall **101** extending between the first side **1000** and the second side **100**D and the first end **100**A and the second end **100**B, a second sidewall **103** also extending between the first side **1000** and the second side **100**D and the first end **100**A and the second end **100**B, an interior cavity **108** within the enclosure **100**, an air supply diffuser **110** disposed at the first end **100**A, and a return air separation baffle **112**. The enclosure **100** includes dimensions that enable the hydronic assembly **68** to fit within cavities having dimensions ranging from (4) inches to ten (10) inches in depth. For example, in embodiments, the enclosure **100** includes a height ranging from fourteen (14) inches to twenty (20) inches, a width ranging from four (4) to six (6) inches, and a length ranging from three (3) feet to five (5) feet.
(59) In some embodiments, the enclosure **100** includes casing angles **105** for mounting the hydronic assembly **68** onto a support structure and for increasing the structural integrity of the hydronic assembly **68** when in use. The casing angles **105** comprise lips **107** extending longitudinally along the length the first sidewall **101** and second sidewall **103**, respectively. The lips **107** are capable of receiving or being mounted onto a support structure such as a bracket or rail. The casing angles **105** define a juncture in which the width of the enclosure **100** tapers to a smaller width. The casing angles **105** increase the rigidity of the hydronic assembly **68** by making the enclosure **100** structurally harder and less prone to rattling and vibrations, which also reduces the generation of noise.
(60) The interior cavity **108** spans the distance between the first end **100**A and the second end **100**B and the distance between the first side **1000** and the second side **100**D. The air supply diffuser **110** includes a return air inlet **110**A for receiving and recirculating air from the unit **33** back into the enclosure **100** and a supply air outlet **110**B for supplying air to the unit **33**. The supply air outlet **110**B may include a longer length, or span a larger area of the first end **100**A, than the return air outlet **110**A. In embodiments, the air supply diffuser **110** is removable from the enclosure **100** so as to detach, disengage, or otherwise separate from the enclosure to enable access to the interior cavity **108** to access all of the components within the enclosure **100** for maintenance and repair tasks. The air supply diffuser **100** may engage the first end **100**A via a friction fit, simply fit over the first end **100**A, or be attached to the first end **100**A via fasteners. In some embodiments, the air supply diffuser **110** may be pivotably connected to the enclosure **100** to enable access to the interior cavity **108**. The air supply diffuser **110** may be pivotably connected to either the first sidewall **101**, the second sidewall **103**, the first side **1000**, or the second side **100**D such that it may open and close with respect thereto. In some embodiments, the air supply diffuser **110** spans the entire first end **100**A such that when removed or open, the air supply diffuser **110** provides an opening spanning the entire first end **100**A to provide full access to the interior cavity **108**. In embodiments, the second end **100**B, the first side **1000**, and the second side **100**D provide no access into the interior cavity **108**, or are closed off, such that the air supply diffuser **110** is the point of access to the interior cavity **108** and the components therewithin.
(61) The return air separation baffle **112** extends from the first end **100**A toward the second end **100**B and separates the return air inlet **110**A and the supply air outlet **110**B. When the air supply diffuser **110** is on or removed from the first end **100**A, the return air separation baffle **110** creates a return air stream section **107** within the interior cavity **108** that is underneath the return air inlet

110A and a supply air stream section **109** within the interior cavity **108** that is underneath the supply air outlet **110**B. In embodiments, the return air separation baffle **112** includes substantially the same width as the enclosure **100** and extends partially across the height of the enclosure **100** to leave a gap **114** between the return air separation baffle **112** and the second end **100**B. The gap **114** enables the air to pass from the return air stream section **107** to the supply air stream section **109**. The return air stream section **107** guides return air **113** through the enclosure **100** past the gap **114** to the supply air stream section **109**. The supply air stream section **109** in turn guides the air toward the hydronic coil **102** and the air supply booster fan **106** such that the air is diffused through the supply air outlet **110**B as supply air **115**. The enclosure **100** includes an opening **111** on the second side **100**D configured to removably receive the branch duct **62** to collect a ventilated air stream from the air duct ventilation system **24**C and mix it with air recirculated into the enclosure **100** through the return air inlet **110**A.

(62) The hydronic coil **102** is disposed within the interior cavity **108** of the enclosure **100** and is removably attachable to the hydronic piping **22**. The hydronic coil **102** is configured to heat or cool air recirculated and ventilated into the enclosure **100**. The hydronic coil **102** includes piping including a finned portion **102**B having a one or more fins **102**C for providing additional heat or cold transfer to air recirculated or ventilated into the enclosure **100**. In embodiments, the fins **102**C are closely spaced metallic projections that enhance the heat transfer from the water circulated through the hydronic coil **102** to the air circulating and mixing within the enclosure **100**. The hydronic coil **102** includes a hydronic coil supply pipe **102**S, which removably attaches to the hydronic supply piping **54**A and a hydronic coil return pipe **102**R, which couples to the hydronic return piping **56**A. In embodiments, each of the hydronic coil supply pipe **102**S and the hydronic coil return pipe **102**R end at the first side **1000** of the enclosure **100** defining an aperture, or junction, that is couplable to the hydronic supply piping **54**A and the hydronic return piping **56**A, respectively. The hydronic coil supply pipe **102**S and the hydronic coil return pipe **102**R are attachable to the hydronic supply piping **54**A and the hydronic return piping **56**A, respectively, via a variety of known pipe fitting pieces such as elbows, tees, reducers, unions, couplings, crosses, caps, swage nipples, plugs, bushings, adapters, outlets, valves, flanges, flexible hoses, braided hoses, and the like. The hydronic coil supply pipe **102**S and the hydronic coil return pipe **102**R each include a section of the finned portion **102**B. The hydronic coil **102** piping is arranged in a series of at least two rows so as to increase the area covered by the hydronic coil **102** within the enclosure **100**.

(63) In embodiments, the hydronic assembly **68** further comprises an air cavity convector **116** removably attachable to the enclosure **100** at the hydronic coil piping **102**. When attached to the enclosure **100**, the air cavity convector **116** is disposed externally with respect to the enclosure **100**, or outside the enclosure **100**. The air cavity convector **116** is configured to deliver heating or cooling to the environment external to the enclosure **100**, such as the air cavity **34** of the unit **33**.

(64) In embodiments, the hydronic coil **102** includes an air cavity convector supply pipe **102**CS and an air cavity convector return pipe **102**CR branching off the hydronic coil **102** piping to the second end **100**B of the enclosure **100**. In some embodiments, the air cavity convector supply pipe **102**CS and the air cavity convector return pipe **102**CR branch off the hydronic soil supply pipe **102**S. The air cavity convector **116** further includes an air cavity convector inlet **116**A removably attachable to the air cavity convector supply pipe **102**CS for receiving water from the hydronic coil **102** and an air cavity convector outlet **116**B removably attachable to the air cavity convector return pipe **102**CR for distributing water back to the hydronic coil **102**. The air cavity convector **116** comprises a finned portion **116**C including one or more fins **116**D to allow for additional heating and cooling transfer to the environment external to the enclosure **100**. In embodiments, the fins **116**D are closely spaced metallic projections that enhance the heat transfer from the water circulated through the air cavity convector **116** to the air cavity **34**.

(65) In embodiments, each of the air cavity convector supply pipe **102**CS and the air cavity

convector return pipe **102**CR end at the second side **100**C of the enclosure **100** defining an aperture, or junction, that is couplable to the air cavity convector inlet **116**A and the air cavity convector outlet **116**B, respectively. The air cavity convector supply pipe **102**CS and the air cavity convector return pipe **102**CR are attachable to the air cavity convector inlet **116**A and the air cavity convector outlet **116**B, respectively, via a variety of known pipe fitting pieces such as elbows, tees, reducers, unions, couplings, crosses, caps, swage nipples, plugs, bushings, adapters, outlets, valves, flanges, flexible hoses, braided hoses, and the like.

(66) The valve control station **104** is disposed within the interior cavity **108** of the enclosure **100**. The valve control station **104** is coupled to the hydronic coil **102** and configured to control the flow of water through the hydronic coil **102**. In embodiments, the valve control station **104** includes an air cavity convector bypass valve **118** configured to selectively shut off the flow of water to the air cavity convector **116**, a shut-off valve **124** to isolate the hydronic coil **102** when needed for maintenance, a control valve **126** to regulate the amount of water flowing through the hydronic coil **102**, and a balancing valve **128** to accommodate pressure variations in the hydronic piping system **22**C. The air cavity convector bypass valve **118** is coupled to the hydronic coil supply pipe **102**S and either of the air cavity convector supply pipe **102**CS or the air cavity convector return pipe **102**CR. The air cavity convector bypass valve may be motorized. In some embodiments, the valve control station **104** includes a strainer to remove particles from the supplied water, and a 4-pipe changeover valve to switch between heating and cooling when both are available simultaneously. In other embodiments, the valve control station **104** includes a small circulation pump instead of the control valve **126** to control the flow of water through the hydronic coil **102**.

(67) The air supply booster fan **106** is disposed within the interior cavity **108** of the enclosure **100**. In embodiments, the air supply booster fan **106** is disposed between the hydronic coil **102** and the air supply diffuser **110**. The air supply booster fan **106** is configured to optimize the supply of air from the enclosure **100** to the unit **33** and mix air recirculated into enclosure **100** with air ventilated into the enclosure **100** prior to the air contacting the hydronic coil **102** to increase efficiency and capacity of the hydronic assembly **68**.

(68) In embodiments, the hydronic assembly **68** includes a filter **120** disposed within the enclosure **100**. The filter **120** is positioned below the return air inlet **110**A between the first side **1000** of the enclosure **100** and the return air separation baffle **112** in order to remove particles from the air returning from the unit **33** to the enclosure **100** through the return air inlet **110**A. In some embodiments, the hydronic assembly **68** includes an auxiliary drip pan **122** disposed at the second end **100**B of the enclosure **100** beneath the hydronic coil **102** to catch condensate dripping from the hydronic coil **102**. The drip pan **122** is disposed between the second side **100**D and the return air separation baffle **112**. The drip pan **122** includes a length greater than or substantially equal to the extent of the finned portion **102**B of the hydronic coil **102** to ensure the drip pan catches all condensate leaking from any part of the finned portion **102**B of the hydronic coil **102**. In some embodiments, the drip pan **122** is disposed between the return air separation baffle **112** and the hydronic coil **102** and includes a length greater than or substantially equal to the height of the finned portion **102**B of the hydronic coil **102**. In some embodiments, the hydronic assembly **68** is configured to operate as a sensible-only cooling device that does not remove moisture from the air in the form of condensation, thereby eliminating the need for a condensate disposal system. However, in embodiments, the auxiliary drip pan **122** may include a leak detector as a safety measure to detect if there is a malfunction and condensation is inadvertently dripping from the hydronic coil **102**. In other embodiments, the hydronic assembly **68** is configured to form condensate and includes a drainage piping system to dispose of the condensate.

(69) As mentioned above, the hydronic assembly **68** is configured to fit recessed within a narrow cavity formed within the air cavity **34** that ranges between 4-10 inches in depth. When a modular panel **16** is installed on a new or existing building to correspond to a particular unit **33**, the only things visible from within the unit **33** are the air supply diffuser **110** and the access door **23** in some

embodiments. The entire hydronic assembly **68** may be removed through the grille **70** when replacement or maintenance is needed.
(70) In operation of embodiments of the present disclosure, air is supplied to the unit **33** through the air supply outlet **110**B, while air is returned from the unit **33** through the adjacent return air inlet **110**A. The air supply booster fan **106** circulates air from the unit **33** through the hydronic coil **102** and supplies the heated or cooled air back into the unit **33**. Within the hydronic assembly **68**, recirculated air is mixed with conditioned ventilation air supplied into the enclosure **100** via the branch duct **62** from the central air distribution system **24**C.
(71) The hydronic supply piping **54**A and hydronic return piping **56**A connect to the hydronic supply riser **54** and the hydronic return riser **56**, respectively, of the central hot and chilled water distribution hydronic piping system **22** to provide the hydronic coil **102** with the necessary water flow for heating and cooling the circulated air. The valve control station **104** includes all components necessary to regulate and control the amount of water flowing through the hydronic coil **102**.
(72) The balancing damper **64** of the branch duct **62** ensures the desired airflow rate of the ventilation air from the air duct **24** is met. The balancing damper **64** can be manual or motorized to allow for adjustment from a digital controller. For example, the balancing damper **64** may control the ventilation airflow rate in response to variables such as the number of occupants within the unit **33** or in response to the dehumidification capacity needed as indicated by a humidity sensor. The balancing damper **64** is accessible through the access door **23**. The access door **23** may be adjacent the air supply diffuser **110**, as shown in FIG. **13**A, to provide access to the balancing damper **64**. The access door **23** provides access to an electrical power and controls box **130** mounted onto the side of the enclosure **100**. In embodiments, the electrical power and controls box may be mounted onto either end **20**A, **20**B or side **20**C, **20**D of the window assembly **20**.
(73) The hydronic assembly **68** includes a temperature setpoint, which may be maintained within the conditioned space by a controller, such as the electrical power and controls box **130**, that adjusts the speed of the air supply booster fan **106** as well as the water flow rate through the hydronic piping **102** based on a signal from a temperature sensor. In some embodiments, the temperature sensor is disposed below the return air inlet **110**A and filter **120** between the first side **1000** and the return air separation baffle **112**. In other embodiments, the temperature sensor is disposed in the unit **33**. Both the air supply booster fan **106** speed and water flow rate through the hydronic coil **102** can be adjusted in discreet steps or modulated continuously within a set range. Individual room controllers may be connected to a central building management system to enable central monitoring and control of the hydronic coil **102**.
(74) Referring now to FIG. **14**A, in some embodiments, the branch duct **62** and the electrical power and controls box **130** are decoupled from the hydronic assembly **68**. The branch duct **62** rather supplies ventilation air directly to the unit **33**. This configuration eliminates the need for access to the balancing damper **64** and electrical power and controls box **130** via the access door **23** next to the air supply diffuser **110**, thereby shortening the length of the overall assembly such that it may fit more easily below a narrow window.
(75) Referring now to FIG. **15**A, in some embodiments, the hydronic assembly **68** is positioned vertically along the second side **20**D of the window assembly **20** to accommodate situations in which there is inadequate vertical clearance below the window assembly **20** for the hydronic assembly **68** and the air cavity convector **116** to fit. In this embodiment, the air cavity convector **116** is decoupled from the enclosure **100** and rather coupled directly to the hydronic supply piping **54**A and hydronic return piping **56**A.
(76) Referring back to FIG. **8**A, FIG. **8**B, and FIG. **8**C, in operation of embodiments of the present disclosure, the ventilation air stream is supplied to each unit **33** at neutral temperature and very low moisture levels to meet any latent cooling loads within the air cavity and/or unit and ensure that the dewpoint within the unit is low enough such that condensation will not form on chilled surfaces.

The air supply/exhaust branch duct **62** delivers a small amount of air directly to the air cavity **34** via the air cavity supply branch **63** for the same purpose of preventing condensation on chilled hydronic piping **22** as well as maintaining a slight positive pressurization within the air cavity **34** to prevent any air infiltration from outdoors.

(77) In operation, the insulated shell formed by the modular attachment of the plurality of modular panels **16** reduces the heating and cooling loads to such low levels that the radiant heating and cooling effect from the air cavities **34** combined with the added capacity produced by the hydronic coil **65** to heat and cool the ventilation air stream are sufficient to meet the entire heating and cooling load within an individual unit **33**.

(78) In embodiments, the anchor **35** is disposed on the upper end **50** of the plurality of modular panels **16**. The anchor **35** extends orthogonally outwardly relative to the interior surface **42** of the plurality of modular panels **16**. In this way, when the anchor **35** attaches to the structure **37** of the building **14**, the plurality of modular panels **16** jut out from the exterior **12** and align parallelly with the exterior **12**, thereby forming the air cavity **34** between the plurality of modular panels **16** and the exterior **12** of the building **14**. In other embodiments, the anchor **35** may comprise a first anchor disposed on the upper end **50** of the plurality of modular panels **16** and a second anchor disposed on the lower end **52** of the plurality of modular panels **16**.

(79) In embodiments, each of the plurality of modular panels **16** comprises an air cavity isolation baffle **72** that separates the air cavity **34** of a modular panel of the plurality of modular panels **16** from the air cavity **34** of an adjacent modular panel of the plurality of modular panels **16**. In effect, the air cavity isolation baffle **72** prevents migration of air, fire, and smoke horizontally or vertically in between adjacent modular panels when the plurality of modular panels **16** are attached to the exterior **12** of the building **14**. The sealant (not shown) is applied around the air cavity isolation baffle **72** to ensure an airtight seal between adjacent modular panels. In some embodiments, the air cavity isolation baffle **72** is disposed between the air cavity supply piping **58** and the second anchor **35**B. In other embodiments, each of the plurality of modular panels **16** comprises an insulated board **77** for mounting the hydronic piping **22** and the air duct **24** onto the plurality of panels **16**.

(80) The present disclosure also provides a method for installing the facade panel conditioning system over the exterior **12** of a new or existing building **14**. In embodiments, the method comprises anchoring the plurality of modular facade panels **16** to the structure **37** of the building **14**, such that each of the plurality of modular facade panels **16** corresponds to the exterior **12** of an individual unit **33** of the building **14** and the air cavity **34** is formed between the exterior **12** of the individual unit **33** and a corresponding modular facade panel. When installed on an existing building, each of the plurality of modular facade panels **16** are anchored to the structure **37** of the building **14** such that the window assembly **20** of a modular panel of the plurality of modular facade panels **16** corresponds in dimension and location to the original window of the individual unit of the building **14** to which the modular panel was anchored.

(81) In embodiments, the method further comprises connecting the hydronic piping **22** of adjacent anchored modular facade panels to form a hydronic piping system that distributes heat throughout air cavities **34** of the plurality of modular facade panels **16** and connecting the air ducts **24** of adjacent anchored modular facade panels to form an air duct ventilation system that distributes air throughout the air cavities **34** of the plurality of modular facade panels **16**. In some embodiments, the method comprises sealing adjacent anchored modular facade panels of the plurality of modular panels **16** to create the insulated shell around the building **14**.

(82) In embodiments, when installed on an existing building, the method further comprises removing the existing windows **31** of the individual units **33** of the building **14** and removing the existing insulation of the interior wall of the individual units **33** of the building **14** that corresponds to the exterior **12** of the individual unit **33**. In some embodiments, the method includes making interior finishes within the individual unit **33** to optimize for radiant heat transfer to the individual unit **33** and thermal comfort within the individual unit **33**. In certain embodiments, the method

includes removing the existing HVAC system components, such as piping, radiators and air conditioning units, from the individual unit **33**.

(83) In embodiments, when installed on a new building, each of the plurality of modular facade panels **16** may include a finished interior wall assembly, including components such as baseboards and electrical outlets to minimize the amount of interior finishing work required after panel installation.

(84) Referring now to FIG. **9** and FIG. **12**, with reference FIG. **8**C, any of the plurality of panels **16** may be prefabricated without the supply riser **54**, the return riser **56**, and the air supply/exhaust riser **60**. Note, some panels of the plurality of panels **16** may be prefabricated as such, while other panels of the plurality of panels **16** are prefabricated in different configurations. For example, in embodiments, the plurality of panels **16** may include only the window assembly **20**, the air cavity supply piping **58**, the air supply/exhaust branch duct **62**, the hydronic assembly **68**, one anchor **35**, and the air cavity isolation baffle **72**. In some embodiments, each of the plurality of panels **16** comprises support framing **17** attached to the interior surface **42** of the panel **16**. The support framing reinforces the structural integrity of the panels **16** and facilitates the mounting of the various components of the panels. In embodiments, the anchor **35** comprises a pair of L-brackets, in which a first L-bracket is disposed on the first side **46** of the panel **16** and a second L-bracket is disposed on the second side **48** of the panel **16**. The air cavity isolation baffle **72** extends across the upper end **50** of the panels **16**. The air cavity isolation baffle **72** may be attached to the anchor **35**. For example, in some embodiments, the air cavity isolation baffle **72** extends from the first L-bracket to the second L-bracket spanning a majority of the length of the upper end **50**.

(85) Note, in the following panel configurations, the panels are demarcated by broken lines. Panel configuration A shows an example panel configuration in which a plurality of panels **16**, not including the supply riser **54**, the return riser **56**, and the air supply/exhaust riser **60** have been modularly attached to the facade of a building. In configuration A, the supply riser **54** and the return riser **56** are provided separately in a supply/return riser panel **19** and the air supply/exhaust riser **60** is provided separately in an air supply/exhaust riser panel **21**. The supply/return riser panel **19** and the air supply riser panel **21** placed in between adjacent panels **16** such that the supply/return riser panel **19** may connect to the air cavity piping **58** of the adjacent panels and the air supply riser panel **21** may connect to the air supply/exhaust branch duct **62** of the adjacent panels **16**. In this way, all modularly attached panels **16** have the piping and ductwork necessary to supply heat and air to the individual units **33**. The supply/return riser panel **19** and the air supply riser panel **21** may attach to the facade of the building itself or to the adjacent panels **16**.

(86) In embodiments, any of the plurality of panels **16** may be prefabricated to either include the supply riser **54**, the return riser **56**, the air supply/exhaust riser **60**, or any combination of the same. Panel configuration B shows an example in which a plurality of panels **16**, some including either the supply riser **54** and the return riser **56** or the air supply/exhaust riser **60**, or all of the same have been modularly attached to the facade of a building. In panel configuration B, a first panel **16**A including the supply riser **54**, the return riser **56**, and the air supply/exhaust riser **60** is positioned between two panels, a second panel **16**B including only the air supply/exhaust riser **60** and a third panel **16**C including only the supply riser **54** and the return riser **56**. Panel **16**A is positioned such that the supply riser **54** and the return riser **56** of panel **16**A is posited adjacent to panel **16**B and the air supply/exhaust riser **60** is positioned adjacent to panel **16**C. In this way, the air cavity supply piping **58** of panel **16**B may connect to the supply riser **54** and the return riser **56** of panel **16**A and the air supply/exhaust branch duct **62** of panel **16**C may connect to the air supply/exhaust riser **60** of panel **16**A to establish the necessary piping and ductwork necessary to provide heat and air to each individual unit **33**. Indeed, individual units **33** may share the piping and ductwork that provides heat and air thereto.

(87) In some embodiments, any of the plurality of panels **16** may be prefabricated to include multiple windows **20**, supply risers **54**, return risers **56**, air supply/exhaust risers **60**, air cavity

supply piping **58**, air supply/exhaust branch ducts **62**, and hydronic assemblies **68**. Panel configuration C shows an example in which a pair of panels **16** including a first panel **16**A having a pair of windows **20**, a pair of supply risers **54**, a pair of return risers **56**, an air supply/exhaust riser **60**, a pair of air cavity supply piping **58**, a pair of air supply/exhaust branch ducts **62**, and a pair of hydronic assemblies **68** and a second panel **16**B a window **20**, an air supply/exhaust riser **60**, air cavity supply piping **58**, an air supply/exhaust branch duct **62**, and a hydronic assembly **68** have been modularly attached to the facade of a building. In panel configuration C, panel **16**A has been configured to span a larger individual unit **33** or two separate individual units **33**. Panel **16**B may be placed on either side of Panel **16**A such that the air cavity supply piping of panel **16**B connects to either of the supply risers **54** or return risers **56** to establish the necessary piping and ductwork necessary to provide heat and air to each individual unit **33**. Panel configuration D shows an example in which one panel **16** includes a plurality of windows **20**, a pair of supply risers **54**, a pair of return risers **56**, a pair of air supply/exhaust riser **60**, a plurality of air cavity supply piping **58**, a plurality of air supply/exhaust branch ducts **62**, and a pair of hydronic assemblies **68**. In panel configuration D, the panel **16** has been configured to span an even larger individual unit **33** or more than two separate individual units **33**. Indeed, depending on the size of the building and the span of the individual units **33** of the building, the plurality of modular panels **16** may be fabricated according to either or any combination of panel configuration A-D.

(88) FIG. **10** shows an alternative embodiment of the panels **16**. The panels **16** in some embodiments, may comprise a vertical rail support system **74** rather than a structural anchor. The vertical rail support system **74** may include a pair of rail support risers **74**A, **74**B extending vertically along the length of the first side **46** and vertically along the length of the second side **48**, respectively, of the panel **16**. The rail support risers **74**A, **74**B may be attached to the panels **16** or come prefabricated as part of the panels **16**. In embodiments, the vertical rail support system **74** is attached to the exterior of the new or existing building in a network of rail support risers extending over the building and then the panels **16** are modularly attached or tied into the vertical rail support risers. The rail support risers **74**A, **74**B may comprise beams such as, I-beams, T-bars, L-angles, wide flange beams, channeled beams, or rectangular beams.

(89) FIG. **11** shows the three-dimensional orientation of a panel **16** with respect to an individual unit **33** of a building **14** when the panel **16** mounted thereon. In embodiments, the anchor **35** attaches to the structure **37** of the building **14** such that the panel **16** juts out from the exterior **12**. The panel **16** spans the height and width of individual unit **33** and/or the area of the exterior **12** of the building **14** that corresponds to the individual unit **33**. The air cavity **34** is formed between panel **16** and the exterior **12** of the building **14**. The hydronic assembly **68** and the air cavity supply piping **58** are positioned within the air cavity **34**. The air supply diffuser **70** of the hydronic assembly **68** spans the area between the window assembly **20** and the windowsill **78** of the individual unit **33** to seal off the unit **33** from the air cavity **34**. The hydronic assembly blows air **80** through the air supply diffuser **70** into the unit **33** to heat or cool the unit **33**. The air cavity supply piping **58** raised the temperature of the air cavity **34**, which in turn raises the temperature of the interior wall **82** of the unit **33**, such that the air cavity supply piping **58** serves as a radiant heat source to the unit **33**.

(90) It is understood that when an element is referred hereinabove as being "on" another element, it can be directly on the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present.

(91) Moreover, any components or materials can be formed from a same, structurally continuous piece or separately fabricated and connected.

(92) It is further understood that, although ordinal terms, such as, "first," "second," "third," are used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These

terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, “a first element,” “component,” “region,” “layer” or “section” discussed below could be termed a second element, component, region, layer or section without departing from the teachings herein.
(93) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, are used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It is understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.
(94) Example embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, example embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein, but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features. Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise shape of a region and are not intended to limit the scope of the present claims.
(95) In conclusion, herein is presented a facade panel conditioning system. The disclosure is illustrated by example in the drawing figures, and throughout the written description. It should be understood that numerous variations are possible, while adhering to the inventive concept. Such variations are contemplated as being a part of the present disclosure.

Claims

1. A hydronic assembly for use with a new or existing building's water distribution system including hydronic piping having a hydronic supply riser and hydronic return riser and for use with the building's air distribution system including an air duct having an air ventilation branch duct to provide heating, cooling, and ventilation to a unit of the building in which it is designed to service, comprising: an enclosure including a first end, a second end, the first end opposite the second end, a first side, a second side, the first side opposite the second side, an interior cavity spanning the distance between the first end and the second end and the distance between the first side and the second side, an air supply diffuser disposed at the first end including a return air inlet for receiving and recirculating air from the unit back into the enclosure and a supply air outlet for supplying air to the unit, the enclosure further including a return air separation baffle extending from the first end toward the second end, the return air separation baffle separating the return air inlet and the supply air outlet; a hydronic coil disposed within the interior cavity of the enclosure, the hydronic coil removably attachable to the hydronic piping and configured to heat or cool air recirculated and ventilated into the enclosure, the hydronic coil including piping and a finned portion including one or more fins to provide additional heat or cold transfer to air recirculated or ventilated into the enclosure, the enclosure removably attachable to the hydronic piping at the hydronic coil piping; a valve control station disposed within the interior cavity of the enclosure, the valve control station coupled to the hydronic coil piping and configured to control the flow of water through the hydronic coil; and an air supply booster fan disposed within the interior cavity of the enclosure between the hydronic coil and the air supply diffuser, the air supply booster fan configured to mix

air recirculated into enclosure with air ventilated into the enclosure prior to the air contacting the hydronic coil.

2. The hydronic assembly of claim 1, wherein: the hydronic coil supply pipe and the hydronic coil return pipe each include a section of the finned portion; and the hydronic coil piping is arranged in a series of at least two rows to increase the area covered by the hydronic coil within the enclosure.

3. The hydronic assembly of claim 1, wherein the enclosure includes an opening on the second side configured to removably receive the ventilation branch duct to collect a ventilated air stream from the air distribution system and mix it with air recirculated into the enclosure through the air return inlet.

4. The hydronic assembly of claim 1, wherein the air supply diffuser is removable from the enclosure, the air supply diffuser spanning the first end such that when open, the air supply diffuser provides an opening spanning the first end that provides access to the interior cavity.

5. The hydronic assembly of claim 1, further comprising a modular panel configured to fixedly attach to a portion of the exterior wall of the new or existing building to enclose the portion of the exterior wall and form an air cavity between the modular panel and the portion of the exterior wall of the building, the modular panel including a window assembly, the hydronic piping, the air duct, an interior surface, an exterior surface, a first side and a second side separated by the window assembly, and an upper end and a lower end separated by the window assembly.

6. The hydronic assembly of claim 5, wherein the window assembly includes an upper end, a lower end, the upper end opposite the lower end, and left side, and a right side, the left side opposite the right side, and an access door.

7. The hydronic assembly of claim 6, wherein the enclosure is removably attachable to the modular panel for easy accessibility and repair.

8. The hydronic assembly of claim 7, wherein to the enclosure is disposed adjacent to the window assembly and protrudes outwardly with respect to the window assembly, the access door positioned adjacent to the ventilation branch duct to access to the ventilation branch duct.

9. The hydronic assembly of claim 8, wherein the window assembly is disposed centrally along a longitudinal axis of the modular panel.

10. The hydronic assembly of claim 1, further comprising a filter disposed below the return air inlet between the first side and the return air separation baffle.

11. The hydronic assembly of claim 10, further comprising a drain pan disposed at the second end of the enclosure beneath the hydronic coil and between the second side and the return air separation baffle, the drain pan including a length greater than or substantially equal to the extent of the finned portion of the hydronic coil.

12. The hydronic assembly of claim 10, further comprising a drain pan between the return air separation baffle and the hydronic coil, the drain pan including a length greater than or substantially equal to the height of the finned portion of the hydronic coil.

13. The hydronic assembly of claim 1, the hydronic coil piping includes a hydronic coil supply pipe and a hydronic coil return pipe, the hydronic coil supply pipe couplable to the supply riser of the hydronic piping and the hydronic coil return pipe couplable to the return riser of the hydronic piping.

14. The hydronic assembly of claim 13, further comprising an air cavity convector removably attachable to the enclosure at the hydronic coil piping, the air cavity convector disposed exteriorly when attached to the enclosure and configured to deliver heating or cooling to the environment external to the enclosure.

15. The hydronic assembly of claim 14, wherein the hydronic coil piping further includes an air cavity convector supply pipe and an air cavity convector return pipe branching to the second end of the enclosure.

16. The hydronic assembly of claim 15, wherein the valve control station includes an air cavity convector bypass valve configured to selectively shut off the flow of water to the air cavity

convector, the bypass valve coupled to the hydronic coil supply pipe and either of the air cavity convector supply pipe or the air cavity convector return pipe.

17. The hydronic assembly of claim 15, wherein the air cavity convector supply pipe and an air cavity convector return pipe branch from the hydronic coil supply pipe.

18. The hydronic assembly of claim 17, wherein the air cavity convector includes an air cavity convector inlet and an air cavity convector outlet, the air cavity convector inlet removably attachable to the air cavity convector supply pipe and the air cavity convector outlet removably attachable to the air cavity convector return pipe.

19. The hydronic assembly of claim 18, wherein the air cavity convector comprises a finned portion including one or more fins to allow for additional heating and cooling transfer to the environment external to the enclosure.